Paleomagnetism and its Relationship to Till Deposition

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Field observations and theoretical studies on temperate glaciers suggest that there is a layer of water at the ice–sediment interface. If this condition existed under continental glaciers it is reasonable to assume that rock and mineral particles were released from the base of the ice through a water or slurry medium.

In order to test this hypothesis, samples of clay till from Port Alma, Ontario, were tested and shown to have a strong component of detrital remanent magnetization (DRM) which resists alternating field demagnetization. The possibility that the remanence alignment was caused by glacier movement has been ruled out by anisotropy of magnetic susceptibility measurements. Thus the results demonstrate that a water or slurry layer existed at the base of continental glaciers at the time of till deposition. The basal water layer may explain the very rapid advance of the glacier lobes through the basins of the Great Lakes. The paleomagnetic data and available radiocarbon dates suggest that the till at Port Alma was deposited in a relatively short period of about 100 to 400 yr. The potential usefulness of DRM in clay-rich tills for stratigraphic studies is being further investigated.

Des observations de terrain et des études théoriques portant sur les glaciers tempérés, suggèrent la présence d'une couche d'eau au contact de la glace et du sédiment. Si cette condition existait pour les glaciers continentaux, il est raisonnable de supposer que des fragments de roches et de minéraux avaient été délogé de la base de la glace par l'intermédiaire d'un milieu fluide (eau vu suspension épaisse).

Dans le but de vérifier cette hypothèse, des échantillon de till argileux provenant de Port Alma, Ontario ont été analysé et ont dévoilé une forte composante de magnétisation rémanente résiduelle (DRM) résistant au champs alternatif de démagnétisation. La possibilité que les mouvements du glacier résultèrent en un alignement des remanences a été écarté dû à l'anisotropie des mesures de susceptibilité magnétique. Donc les résultats démontrent qu'une couche d'eau ou de fluide à suspension épaisse existait à la base des glaciers continentaux durant la sédimentation du till. Cette couche d'eau peut expliquer l'avancée rapide des lobes du glacier via les bassins des Grands Lacs. Les données paléomagnétiques et les dates au radiocarbone disponibles suggèrent que le till de Port Alma a été déposé dans un laps de temps relativement court, de l'ordre de 100 à 400 ans. L'utilité relative du DRM, à des fins d'études stratigraphiques, dans les tills argileux est également discutés. [Traduit par le journal]

Introduction

There is general agreement among glacial geologists that till is deposited from glaciers in two distinct ways. First, there is till that is deposited from the base of the glacier and is commonly referred to as "basal till" (*e.g.* Goldthwait 1971, p. 15). Alternatively, material which accumulated on the surface of a glacier and was let down from the surface as the ice melted is referred to as "ablation till". The present discussion deals with some aspects of the deposition of basal till.

There is some disagreement on the method by which particles are released from the base of a glacier. In general, most authors refer to basal till as being deposited from the base of moving ice by a "plastering on" effect (*e.g.* Holmes 1941, p. 1312, Flint 1971, p. 171), whereas others (*e.g.* Harrison 1957, p. 301) consider that basal till was deposited by melting at the base of stagnant ice. In both cases, however, it is usually assumed that the particles were released from the ice by basal melting, using geothermal and frictional heat which may

be augmented by heat from meltwaters percolating down through the ice, or a combination of these phenomena.

Thus it is reasonable to assume that at the time of release of material from the base of the ice, the rock and mineral particles would fall through a water layer or a slurry composed of water and suspended rock and mineral fragments.

Following on this assumption it is possible that very small magnetic particles released from the base of the ice would have the opportunity to rotate and align themselves in the earth's magnetic field while passing through the water or slurry medium. The thickness of the water or slurry medium required for such alignment is not known, but experiments on the deposition of artificial sediments (Doell and Cox 1961, p. 233) show that only a small amount of water in the order of 50% by weight of the total slurry is necessary for magnetic particles to align themselves during deposition. It is possible that some rotation of the magnetic particles would continue at the base of the water or slurry layer, but once the particles are buried by a few centimeters of clay-rich unsorted materials it is unlikely that any further rotation would take place.

In order to test this hypothesis a section of till from Port Alma, Ontario (lat. 42°07′ N, long. 82°07′ W) on the north shore of Lake Erie was sampled (Fig. 1). The bluffs at this location are about 12 m in height and composed of a uniform massive unweathered clay till. This same till sheet is exposed for several kilometers west of the Port Alma site and at least 70 km east of Port Alma. Examination of the till at 20 sites along the north shore of Lake Erie shows that in all cases the till is massive in character and lacks evidence of stratification or interlayered stratified sediments, which suggests that the till is a 'normal' basal till and not deposited from the base of an ice shelf. This till sheet is of late Wisconsin age and named the Port Stanley Till (De Vries and Dreimanis 1960). A granulometric analysis of 2000 g of the till at Port Alma shows that it contains 8.5% + 1 mm, 19.5 % − 1 mm + 63μ, 28% − 63μ + 4μ and 44% − 4μ.

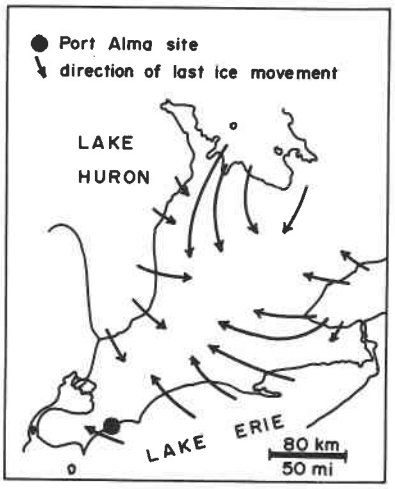


FIG. 1. Map of southwestern Ontario showing the location of Port Alma and the directions of last ice movement (after Chapman and Putnam 1951, and Dreimanis 1961).

Sampling Procedure

The best method of sampling found to date has been to cut oriented blocks of till measuring approximately 25 cm × 12 cm × 12 cm, from which an average of five cores were obtained by placing the block in a drill press and coring with a diamond bit. The cores were then carefully trimmed to size, 2.5 cm in diameter and 2.5 cm in length, labelled, and coated with polyurethane paint. The polyurethane paint did not soak into the samples and provided an excellent tough transparent coating.

Because the till bluffs are nearly vertical in the Port Alma area, it proved difficult to cut out oriented samples throughout the entire section. Accordingly the bluffs were sampled at 25 cm intervals from water level to 2.8 m above water level, and from 7.4 m above water level to 9.9 m above water level. Two other sites in the same area were also sampled and gave similar results to those reported in this paper.

Remanent Magnetization

The natural remanent magnetization (NRM) of one specimen from each block was measured initially using a Schonstedt SSM-1A spinner magnetometer. The remanence was remeasured after each cleaning treatment, using a Schonstedt GSD-1 demagnetizer employing alternating field (a.f.) demagnetization in fields of 50, 100, 200, 400, and 500 Oe peak intensity. Several specimens were cleaned at smaller field increments and up to 800 Oe. All specimens

TABLE 1. Summary of remanent magnetization data

Height of block above lake level (meters)	Number of specimens	Length of vector resultant R	Mean remanence direction				Virtual North Pole position			
			Declination (degrees)	Inclination (degrees)	Precision parameter k	α_{95} (degrees)	Longitude West (degrees)	Latitude North (degrees)	δ_p (degrees)	δ_m (degrees)
0	5	4.9657	9.7	45.9	116	7.1	293.3	73.2	5.8	9.1
0.25	4	3.9753	10.5	42.6	121	8.4	291.8	70.5	6.4	10.3
0.50	6	5.9830	14.2	43.3	295	3.9	301.5	69.5	3.0	4.9
0.75	4	3.9777	37.2	45.0	135	8.0	336.9	55.9	6.4	10.1
1.25	3	2.9866	18.9	57.7	149	10.1	343.9	75.1	10.9	14.9
1.50	4	3.9644	359.7	44.6	84	10.1	261.0	74.1	7.9	12.7
1.75	4	3.9883	11.7	43.0	256	5.8	295.1	70.3	4.4	7.1
2.00	3	2.9125	22.9	46.8	23	26.4	322.4	66.7	21.9	34.0
2.25	4	3.9641	10.7	50.0	83	10.1	302.6	75.8	9.03	13.5
2.50	4	3.7255	1.5	41.2	11	29.1	266.4	71.5	21.6	35.5
2.75	3	2.9877	14.5	34.3	162	9.7	294.3	63.7	6.4	11.1
5.60	7	6.9447	13.3	45.1	108	5.8	301.4	71.1	4.7	7.4
7.40	5	4.9920	16.4	32.3	499	3.4	296.8	61.7	2.2	3.9
8.10	5	4.9850	6.1	39.1	268	4.7	278.4	69.4	3.3	5.6
8.40	6	5.9601	342.3	38.9	128	5.9	220.2	65.0	4.2	7.1
8.90	7	6.9556	2.3	33.4	135	5.2	267.5	66.1	3.4	5.9
9.40	7	6.9638	3.8	48.6	166	4.7	277.0	77.0	4.1	6.2
9.65	5	4.9810	358.7	35.4	210	5.3	258.9	67.4	3.5	6.1
9.90	4	3.9902	14.0	33.1	305	5.3	292.8	63.1	3.4	6.0
Mean										
By specimen	90	87.8052	8.4	42.3	41	2.4	286.3	71.0	1.8	2.9
By block	19	18.6933	9.6	42.6	59	4.4	289.7	70.9	3.4	5.5

NOTES: α_{95} = radius of cone of 95% confidence, δ_p, δ_m = semi-axes of oval of 95% confidence along and perpendicular to the site-pole great circle.

reacted to demagnetization similarly. Their remanence intensity (Fig. 2) decreased nearly linearly with increasing a.f. intensity, giving a mean coercivity at about 300 Oe (*i.e.*, a.f. required to reduce NRM intensity by one half). Their remanence directions typically show a few degrees (~4°) of movement away from the earth's present magnetic field direction up to 50 Oe, presumably reflecting the removal of viscous remanent magnetization (VRM) components, a clustering between 50 and 400 Oe which is thought to be the characteristic remanent magnetization (RM_c), and an increasing scatter about the cluster above 400 Oe as isothermal remanence components are induced in the specimens by the demagnetizing process (Fig. 3). A storage test of one month duration revealed that the till specimens acquire relatively small VRM components (VRM $\leqslant$ 0.03 NRM), which are removed by 100 Oe a.f. cleaning. On the basis of these cleaning tests, the remaining specimens were a.f. cleaned at 200 Oe and their RM_c measured. Assigning the RM_c direction of each of the three to seven specimens per block unit weight, the mean characteristic remanent magnetization (RM_m) direction was calculated for each block along

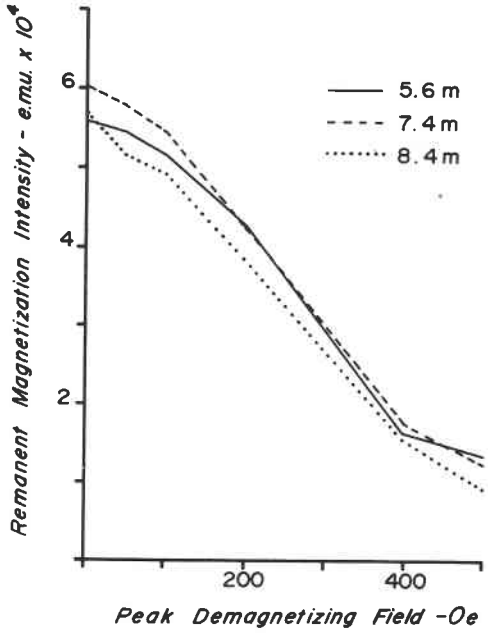


FIG. 2. Change in remanent magnetization intensity on a.f. step demagnetization of the same three example till specimens as in Fig. 3.

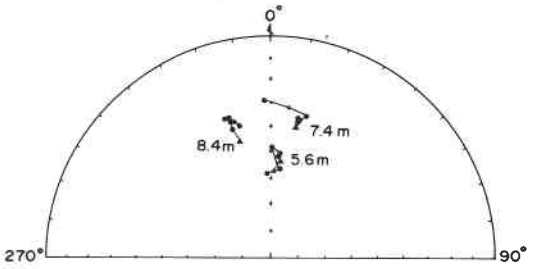


FIG. 3. Change in remanent magnetization direction on a.f. step demagnetization of three example till specimens. From the NRM direction (solid triangle), the remanence goes progressively to the direction after 50, 100, 200, 400, and 800 Oe cleaning. The number indicates the height of the till block above lake level in meters.

with its precision parameter (k), radius of the cone of 95% confidence (α_{95}), virtual geomagnetic pole position, and semi-axes (δ_p, δ_m) of the oval of 95% confidence about the pole (Fisher 1953) (Table 1).

Discussion of Remanence Results

The within-block variations in RM_c directions are small, resulting in RM_m directions with high k and low α_{95} values. Thus the Port Stanley Till exhibits a statistically significant variation in RM_m directions at different elevations in the section.

It is therefore deduced for the Port Stanley Till that material was released from the base of the ice by melting so that the particles were deposited from a water slurry, thereby allowing the magnetite particles to rotate into alignment with the earth's magnetic field to form a detrital remanent magnetization (DRM) in the till.

The variations in declination and inclination are shown in Figs. 4 and 5. They exhibit a 'cyclic' behavior for approximately every 125 cm of till section, during which the declination oscillates about 30° and the inclination about 15°. When the virtual geomagnetic poles (VGP) calculated from the RM_m directions using the dipole hypothesis for these till blocks are examined (Fig. 6), it is apparent that they cluster in north-central Russia. Relative to the great circle passing through the sampling site and geographic pole, the calculated mean paleomagnetic pole is far-sided by 19° and right-handed by 9°. The 3° far-sided and 3° right-handed bias of the earth's magnetic pole found by Wilson (1971) for lavas in the last 25 m.y.

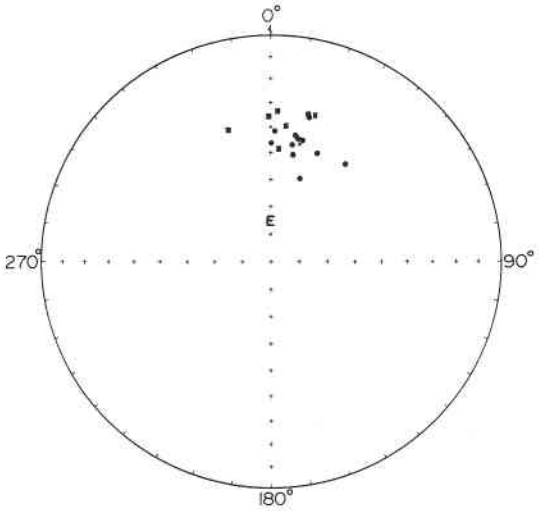


FIG. 4. Stereographic projection of block mean remanent magnetization directions. The solid circles represent blocks from 0 to 110 in. (0–2.79 m) above lake level, the × from 220 in. (5.59 m), and the solid squares from 290 to 390 in. (7.37–9.91 m). The E indicates the Earth's present magnetic field direction.

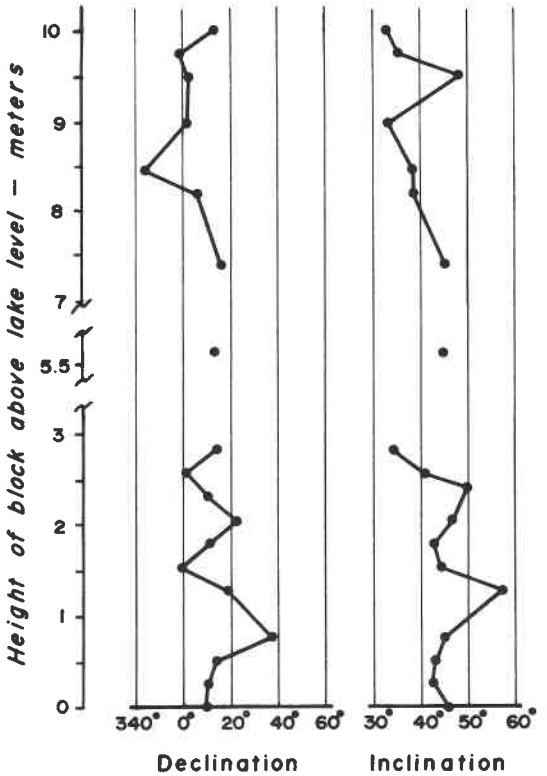


FIG. 5. Variation in average remanent magnetization directions of blocks of till against height above lake level in the Port Alma section.

may contribute slightly to this result, but cannot explain it. If the till's DRM accurately records the earth's magnetic field, the 'cyclic' behavior and mean pole reflect the meandering movement of the magnetic pole during a portion of a secular variation cycle. The secular variation periodicity is known to be in the order of 10^3 yr (Cox 1968). The RM_m directions would then suggest that the 9.9 m sample section of the Port Stanley Till was deposited rapidly during an interval of 100 to 400 yr. If, on the other hand, the alignment of the magnetic particles has been biased during deposition by the action of glacier or of the sedimentation process, then the 'cyclic' behavior could reflect several (*i.e.*, $\geqslant 4$) secular variation cycles about a far-sided mean pole displaced by flattening of the remanence inclination. The immediate criticism of this alternative is that, as will be further discussed, available radiometric age dates and other geologic evidence suggest that the Port Stanley Till was deposited during a relatively short period.

Magnetic Mineralogy

Linear decay of remanence intensity on a.f. step demagnetization is typical of rocks with a small effective magnetic domain size caused commonly by exsolution structures in the (titano) magnetite (Strangway 1970, p. 79). The linear decay curves found for the Port Stanley Till are caused simply by the small magnetite particle size.

In order to determine the particle size of the magnetite in the till, a 2000 g sample was sieved through a 1 mm opening screen to remove the coarse fraction. The −1 mm material was then screened through a series of sieves down to 400 mesh (37μ). Fractions remaining on each sieve were weighed and then split down to 25 g samples, which were placed in a heavy liquid (tetrabromoethane; Sp.Gr. 2.96) and the heavy minerals separated from the light fraction. The magnetite was then separated from the heavy minerals using a strong hand magnet, and the magnetite so obtained was weighed and examined under a microscope to determine the number of non-magnetic particles present. In general it was found that less than 5% of the magnetically

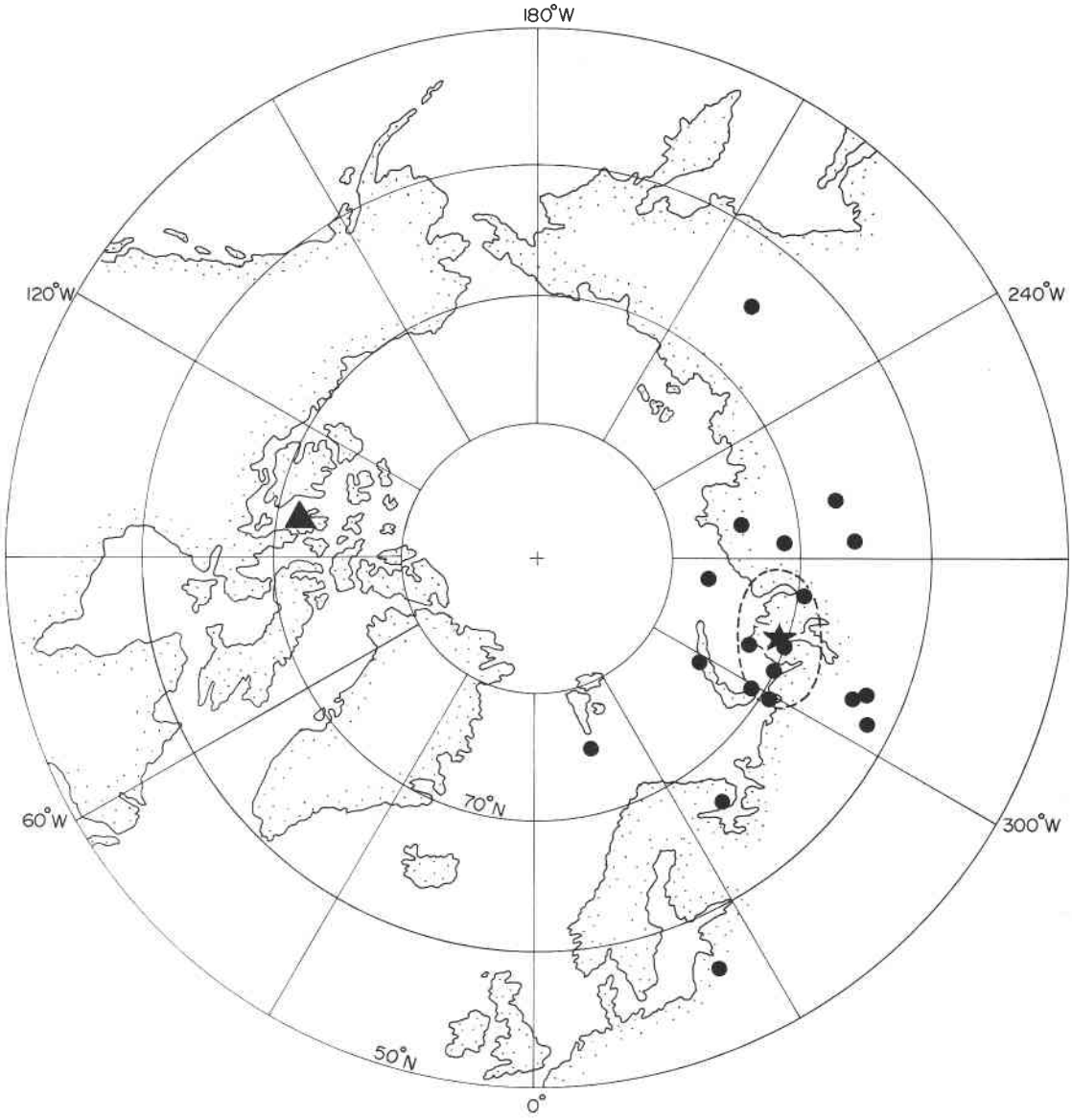


FIG. 6. Polar projection north of latitude 50° N showing the pole positions derived from the till block mean remanent magnetization directions (solid circles), with their mean position (star), and its oval of 95% confidence (dashed line). The present north magnetic pole is indicated by a solid triangle.

separated material is composed of non-magnetic material.

Attempts to use the same method of separation on the remaining silt and upper clay fraction (*i.e.*, $37\mu - 1\mu$) proved unsuccessful, as the very small heavy minerals were trapped and floated with the light fraction. In order to overcome this problem a simple technique was devised which consisted of making a slurry of the silt fraction in a beaker and then placing the beaker on a commercial magnetic stirrer with the plastic-coated stirrer in the slurry. The stirrer was weighed before and after immersion and in this way the weight of the magnetite in this fraction was determined. Microscopic examination revealed that less than 5% of the

TABLE 2. Data from anisotropy of magnetic susceptibility

Height of block above lake (meters)	Number of specimens	*Maximum axis*			*Minimum axis*			*Mean axial magnitude ratios*	
		Declination (degrees)	Inclination (degrees)	α_{95} (degrees)	Declination (degrees)	Inclination (degrees)	α_{95} (degrees)	Max/Int	Min/Int
0.25	7	327.7	10.3	31.2	89.0	78.7	7.5	1.017	0.915
1.50	3	342.3	8.8	87.6	101.2	79.9	5.6	1.013	0.886
1.75	4	20.6	7.6	50.8	77.6	88.7	2.8	1.022	0.873
2.25	4	323.1	5.6	24.6	86.0	88.0	8.7	1.034	0.920
5.60	7	22.9	17.8	32.2	163.9	81.9	6.4	1.014	0.926
8.40	6	321.0	9.8	16.1	146.0	84.5	8.1	1.039	0.910
Average	6*	345.5	11.1	24.5	114.7	84.5	4.6	1.025	0.905

*Number of blocks.

silt-sized material was non-magnetic and the average size was about 20μ with some particles less than 1μ in size. The length and width of 75 particles in each size range were measured and each case gave an average length-to-width ratio of greater than 1.5 : 1.

By knowing the weight and average length and width of particles in each fraction it is possible to determine the number of particles in each size range. The computations show that there are approximately 10^5 magnetite particles/cm^3 of till in the minus 37μ range, about 150 particles/cm^3 in the 37μ – 63μ range, and another 150 particles/cm^3 in the plus 63μ range. This large number of small particles of magnetite in the till specimens is consistent with the findings of a relatively large bulk magnetic susceptibility, which averages 6.5×10^{-5} emu/cm^3 throughout the till section combined with a low within-block susceptibility variation ($\approx$0.5 to 1.0×10^{-5} emu). The bulk susceptibilities were measured on a toroid transformer bridge apparatus (Christie and Symons 1969).

Anisotropy

Although the remanent magnetism strongly suggests that the magnetic particles were aligned in the earth's field during deposition of the till, there remains the possibility that this alignment was developed primarily or biased by the action of glacier flow. It is generally agreed that the long axes of clasts in till are statistically aligned parallel to the direction of ice movement (Embleton and King 1971, p. 307). Consequently it would be expected that the long axes of the magnetic particles would also be aligned parallel to the direction of ice flow. This possibility has been explored by Fuller (1964), who used magnetic susceptibility anisotropy as a rapid method for fabric determination. He showed that the magnetic anisotropy method in tills provides similar results to those obtained from conventional methods of 3-dimensional fabric analysis. Utilizing the 6-spin method on the Schonstedt SSM-1A Magnetometer and the bulk susceptibility measured in the K_{22} direction on the transformer bridge, the ellipsoid of anisotropy of magnetic susceptibility was measured for 31 specimens from 6 till blocks at various heights in the section.

For each specimen the magnitude and direction of the principal susceptibilities were calculated by the method of successive approximations (Nye 1960, p. 165). The results are illustrated in Fig. 7 and summarized in Table 2. The minimum susceptibilities are obviously tightly clustered close to the vertical direction. This means that the plane of maximum and intermediate susceptibility is nearly horizontal, and that the long axes of the magnetite particles are preferentially biased towards the horizontal plane. The clustering of the maximum and intermediate susceptibilities is not as tight as for the minimum susceptibilities. This is typical, however, of sedimentary fabrics in which the maximum and intermediate susceptibility intensities differ by about 1% and the minimum differs from both by a much larger percentage (Stone 1963, p. 385). It is notable that the maximum susceptibilities are biased into the northwest quadrant as well as into the horizontal plane. The preferential alignment of the maximum susceptibility indicates that the long axes of the magnetite particles are preferentially directed towards the northwest. This bias is thought to have originated by the magnetite particles being aligned parallel to the flow of the ice while entrapped in the ice and retaining this alignment on being released into the fluid slurry and dropping through it. This hypothesis is consistent with the coincidence of the maximum susceptibility direction and the known direction of the last ice movement in the area.

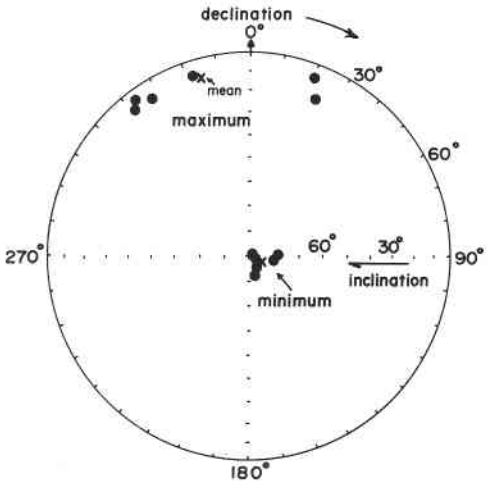


FIG. 7. Stereographic plot of directions of maximum and minimum susceptibility. Solid (open) circles indicate downward (upward) directed inclinations.

The general geologic setting (Fig. 1) favors a northwesterly ice movement direction as the glacial ice moved in a westerly direction through the Erie basin and spread out to the north where the Erie lobe met ice coming down the Huron basin. The long axes of end moraines found north of the Erie basin in southwestern Ontario confirm this observation (Chapman and Putnam 1951, Dreimanis 1961). The slight statistically significant dip of the maximum susceptibility axis towards the northwest is evident although its cause is not.

Three factors strongly suggest that the RM_c directions in the till have not been biased by the glacial flow action. First, the RM_c directions are moderately steeply inclined with respect to the maximum-intermediate susceptibility plane, which is not very indicative of biasing. Second, the percentage of magnetite in the till is sufficiently low, (*i.e.* $\leqslant 1\%$) that the RM_c intensities and susceptibility magnitudes are weak compared to the earth's magnetic field intensity. Thus the biased alignment of magnetite grains by glacial flow would not interact magnetically to alter the earth's magnetic field direction and thereby bias the alignment of subsequent magnetite grains in producing their RM_c directions. Third, the axial ratios for the anisotropy of susceptibility ellipsoid are relatively close to 1.0 (Table 2). From experiments the maximum biasing of the RM_c directions into the maximum-intermediate susceptibility plane would be 3° under optimum conditions (Nagata 1962). Considering all these factors it seems unlikely that the action of glacial flow has altered or biased the RM_c directions in the till by as much as 1°.

Inclination Error

When sediment grains settle through a water layer, the depositional DRM of the sediment agrees in declination with the applied magnetic field, but its inclination is often less than the field inclination giving an 'inclination error'. Inclination errors of up to 25° have been reported in laboratory sedimentation experiments, with the magnitude dependent on factors such as grain size, grain shape, water flow velocity, and water depth (Irving 1964, p. 30). An inclination error of 25° could give a 35° far-sided pole position and hence is cause for concern. Unfortunately most experiments have been carried out on well-sorted particle sizes (*e.g.*, Nagata 1962) under limited hydrostatic heads and therefore are not particularly applicable to poorly sorted tills deposited under potentially large static heads. Also, if particles are released in a thin layer of slurry, as seems likely, the DRM may be mostly postdepositional, *i.e.*, acquired by particle rotation after deposition but before consolidation. In this case the DRM would be aligned parallel to the applied magnetic field (Irving and Major 1964; Stacey 1972). In addition oceanic sediments deposited under large hydrostatic heads do not show inclination errors (Opdyke and Henry 1969).

In order to estimate the magnitude of the potential inclination error in till, several samples of till from Port Alma were mixed with various amounts of water and then dried in the earth's magnetic field by applying a vacuum to the underside of the sample container. The results showed that in the case of deposition from a thick slurry—the order of 50% by weight water—the inclination error is negligible. With increasing water content, however, the error increased up to 30°. It is considered that not too much significance should be attached to these very preliminary experiments as it is difficult to simulate conditions under a glacier and most of the inclination error developed in 'thin' slurries is likely due to the non-uniform shrinkage of the till in the drying process. As an initial generalization it appears unlikely that inclination error is responsible for more than about 5° of the 19° far-sided pole positions found for the Port Alma till; and further it cannot account for the right-handedness of the position. Further studies on this problem are currently in progress.

Rate of Deposition of the Port Stanley Till

The time of advance and retreat of the ice which deposited the Port Stanley Till is reasonably well documented by radiocarbon dates (Goldthwait *et al.* 1965, p. 94). The advance of the glacier started in the eastern end of the Erie basin about 14 700 yr B.P. It moved in a southwesterly direction to its limit of advance, which is marked by the Powell moraine in Ohio. After the deposition of the Powell moraine the ice retreated to the northeast back through the

Erie basin well to the east of the Port Alma site. The retreat ended about 13 500 yr B.P. Thus the available data indicate that the Port Stanley Till was deposited during this 1200-yr time interval.

The cyclical nature of the variations of the average remanent magnetization of blocks of till against height of till above lake level (Fig. 5) would suggest that the till was deposited at a constant rate. Thus it may be assumed that the rate of advance and retreat of the ice which deposited the Port Stanley Till was reasonably uniform. Therefore the time taken for the ice to advance from the eastern end of the Erie basin to the Port Alma site and retreat from the Port Alma site to the eastern end of the Erie basin can be calculated and subtracted from the 1200 yr time interval to give the time that the Port Alma site was ice-covered. Taking this factor into account, it is likely that the Port Alma site was ice-covered for the order of 600 to 800 yr.

In addition, the 12 m section of till at Port Alma is not the total thickness of the Port Stanley Till at this site. The till extends below the level of Lake Erie and its top was planed off by wave action in glacial lakes that covered the area after the ice retreated east of Port Alma. At the top of the section there is a substantial thickness (the order of 1.6 to 3 m) of sand and gravel. This sand and gravel has been opened up at several locations for use as aggregate and its bedding is typical of beach and related near-shore deposits. Thus it is reasonable to assume that this sand and gravel was derived from reworking the upper till by wave action. From the granulometric analysis of the till this thickness of sand and gravel would represent a removal of from 5 to 10 m of till. Thus the total thickness of the till was probably a minimum of 17 m and possibly closer to 19 to 25 m. Thus the 12 m of till at Port Alma was deposited during a time interval of approximately 400 to 500 yr to give an average rate of deposition of about 3 cm yr^{-1}.

This rate of deposition fits reasonably well with the paleomagnetic data, which showed that the RM_m poles tend to cluster on the 'far side' of the earth's rotational pole and the deduction from paleomagnetic data that the Port Alma till section was deposited during a short time interval, the order of 100 to 400 yr.

Conclusions

Measurements on the Port Stanley Till show that this clay-rich till has a characteristic remanent magnetism (RM_c) which was developed when the magnetic particles were released through a water or slurry layer at the base of the ice, thus forming a detrital remanent magnetization (DRM). The possibility that the RM_c directions were developed or biased by glacier motion has been ruled out by magnetic susceptibility anisotropy, which shows that the long axes of the magnetic particles are shallowly inclined in a northwest direction and bear no relationship to the steeply inclined RM_c directions.

These results open up the possibility that other till sheets may also have an RM_c which could be used for till correlation studies. It must be kept in mind, however, that the Port Stanley Till is ideal in that it can be readily cored and has a very large number of magnetic particles in the fine silt fractions. In addition it is recognized that till deposition is time-transgressive and hence long-range correlations should not be attempted before more field studies are made on single till sheets which can be traced over long distances.

Another use that may be made of the technique is to differentiate superimposed till sheets where the relative age of the till is in question. For example, there are many outcrops which show several till sheets where there is no evidence of weathering between the deposits. In some instances there may be major time breaks between the tills and in other instances the tills may simply represent minor fluctuations of the ice front. In many such cases, color, fabric, lithology etc. are used to determine the differences in the till. While these techniques are all useful an element of doubt sometimes remains. In such cases, if the tills have an RM_c it might be possible to quickly distinguish those tills which were deposited during a short time interval (*i.e.*, representing ice front fluctuations) versus those where a major time break is present but not represented by a buried soil profile or other evidence of weathering.

In the context of till deposition the main contribution of this study is that there must have been a water or slurry layer at the base of continental glaciers to allow the magnetic particles to become oriented in the earth's field

during deposition. The presence of this water layer and its lubricating effect may explain the rapid advance of the ice which penetrated through the Great Lakes basins and into the north-central parts of the United States.

Next the cyclical nature of the RM_c against depth of till suggests that during an ice lobe advance the rate of deposition of till is reasonably constant. The fact that the section of till at Port Alma was probably deposited during only a part of a secular variation cycle indicates that the 12 m section of till was likely deposited in a short time interval of 100 to 400 yr. This result is consistent with available data obtained from radiocarbon dates on the advance and retreat of the ice that deposited the Port Stanley Till.

It is evident that future studies on the magnetic properties of till will yield useful information on the stratigraphy and genesis of this, the most common of glacial deposits.

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CHAPMAN, L. J. and PUTNAM, D. F. 1951. The physiography of Southern Ontario. Univ. Toronto Press, Toronto, Ont., 284 p.

CHRISTIE, K. W. and SYMONS, D. T. A. 1969. Apparatus for measuring magnetic susceptibility and its anisotropy. Geol. Surv. Can., Pap. 69–41, 10 p.

COX, A. 1968. Lengths of geomagnetic polarity intervals. J. Geophys. Res. **73**, pp. 3247–3260.

DEVRIES, H. and DREIMANIS, A. 1960. Finite radiocarbon dates of the Port Talbot interstadial deposits in Southern Ontario. Science **131**, pp. 1738–1739.

DOELL, R. R. and COX, A. 1961. Paleomagnetism. *In*: Advances in geophysics, (*Eds.*), H. F. Landsberg and J. Van Mieghem, Academic Press, **8**, pp. 221–313.

DREIMANIS, A. 1961. Tills of Southern Ontario. *In*: Soils in Canada, R. F. Legget (*Ed.*), Univ. Toronto Press, Toronto, Ont., pp. 80–96.

EMBLETON, C. and KING, C. A. M. 1971. Glacial and periglacial geomorphology. Macmillan Co. Can. Ltd., Toronto, Ont., 608 p.

FISHER, R. A. 1953. Dispersion on a sphere. Roy. Soc. London, Proceed. Ser. **A 217**, pp. 295–305.

FLINT, R. F. 1971. Glacial and Quaternary geology. John Wiley and Sons, N.Y., 892 p.

FULLER, M. D. 1964. A magnetic fabric in till. Geol. Mag. **99**, pp. 233–237.

GOLDTHWAIT, R. P. *et al.* 1965. Pleistocene deposits of the Erie lobe. *In*: The Quaternary of the United States, H. E. Wright and D. G. Frey (*Eds.*), Princeton Univ. Press, Princeton, N.J., pp. 85–97.

GOLDTHWAIT, R. P. 1971. Introduction to till, today. *In*: Till: A symposium, R. P. Goldthwait (*Ed.*), Ohio State Univ. Press, Columbus, Ohio, pp. 3–26.

HARRISON, P. W. 1957. A clay till fabric: its character and origin. J. Geol. **65**, pp. 275–304.

HOLMES, C. D. 1941. Till fabric. Bull. Geol. Soc. Am., **52**, pp. 1299–1354.

IRVING, E. 1964. Paleomagnetism. John Wiley and Sons, N.Y., 399 p.

IRVING, E. and MAJOR, A. 1964. Post-depositional detrital remanent magnetism in a synthetic sediment. Sedimentology **3**, pp. 135–143.

NAGATA, T. 1962. Notes on the detrital remanent magnetization of sediments. J. Geomagn. Geoelec., **14**, pp. 99–106.

NYE, J. F. 1960. Physical properties of crystals. Clarendon Press, Oxford, 322 p.

OPDYKE, N. D. and HENRY, K. W. 1969. A test of the dipole hypothesis. Earth Planet. Sci. Lett. **6**, pp. 139–151.

STACEY, F. D. 1972. On the role of Brownian motion in the control of detrital magnetization of sediments. Pure Appl. Geophys. **98**, pp. 139–145.

STONE, D. B. 1963. Anisotropic magnetic susceptibility measurements on a phonolite and on a folded metamorphic rock. Roy. Astron. Soc., Geophys. J. **7**, pp. 375–390.

STRANGWAY, D. W. 1970. History of the earth's magnetic field. McGraw-Hill Book Co., N.Y., 168 p.

WILSON, R. L. 1971. Dipole offset—the time-average palaeomagnetic field over the past 25 million years. Geophys. J. Roy. Astron. Soc. **22**, pp. 491–504.